

# Jaesung Rim

Ph.D.

POSTECH

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Webpage: <https://rimchang.github.io>

## Education

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| Sep. 2020 ~<br>Feb. 2026 | <b>Pohang University of Science and Technology</b><br>Ph.D. in Graduate School of Artificial Intelligence<br><i>Advisor: Sunghyun Cho</i>                                                                                                 | Pohang,<br>Korea |
| Sep. 2018 ~<br>Aug. 2020 | <b>Daegu Gyeongbuk Institute of Science and Technology</b><br>M.S. in Information and Communication Engineering<br>Thesis: Real-World Blur Dataset for Learning and Benchmarking<br>Deblurring Algorithms<br><i>Advisor: Sunghyun Cho</i> | Daegu,<br>Korea  |
| Mar. 2011 ~<br>Aug. 2017 | <b>Kwangwoon University</b><br>B.S. in Industrial Psychology                                                                                                                                                                              | Seoul,<br>Korea  |

## Experience

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| Mar. 2026 ~<br>Present   | <b>Huawei Singapore Research Center</b><br>AI Engineer at Camera ISP Team            | Singapore           |
| May. 2024 ~<br>Aug. 2024 | <b>Huawei Finland Research Center</b><br>Research Intern at Camera AI Solutions Team | Tampere,<br>Finland |

## Research Interests

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- Computational photography
- Image/video restoration
- Deblurring
- Mobile imaging

## Publications

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1. **Jaesung Rim**, Woohyeok Kim, Haeyun Lee, Heemin Yang, Ke Wang, Sunghyun Cho, "Gyro-based Deep Video Deblurring", CVPR 2026
2. Woohyeok Kim, **Jaesung Rim**, Daeyeon Kim, Sunghyun Cho, "Dynamic Exposure Burst Image Restoration", CVPR 2026
3. Heemin Yang, **Jaesung Rim**, Seungyong Lee, Seung-Hwan Baek, Sunghyun Cho, "Gyro-based neural single image deblurring", CVPR 2025

4. **Jaesung Rim**, Junyong Lee, Heemin Yang, Sunghyun Cho, "Deep Hybrid Camera Deblurring for Smartphone Cameras", SIGGRAPH 2024
5. Sanghyun Kim\*, Min Jung Lee\*, Woohyeok Kim, Deunsol Jung, **Jaesung Rim**, Sunghyun Cho, Minsu Cho, "Burst Image Super-Resolution with Base Frame Selection", CVPRW NTIRE 2024
6. Dongwoo Lee, Jeongtaek Oh, **Jaesung Rim**, Sunghyun Cho, Kyoung Mu Lee, "ExBluRF: Efficient Radiance Fields for Extreme Motion Blurred Images", ICCV 2023
7. Sohyun Lee\*, **Jaesung Rim**\*, Boseung Jeong, Geonu Kim, ByungJu Woo, Haechan Lee, Sunghyun Cho, Suha Kwak, "Human Pose Estimation in Extremely Low-Light Conditions", CVPR 2023
8. **Jaesung Rim**, Geonung Kim, Jungeon Kim, Junyong Lee, Seungyong Lee, Sunghyun Cho, "Realistic Blur Synthesis for Learning Image Deblurring", ECCV 2022
9. Junyong Lee, Hyeongseok Son, **Jaesung Rim**, Sunghyun Cho, Seungyong Lee, "Iterative Filter Adaptive Network for Single Image Defocus Deblurring.", CVPR 2021
10. **Jaesung Rim**, Haeyun Lee, Jucheol Won, Sunghyun Cho, "Real-World Blur Dataset for Learning and Benchmarking Deblurring Algorithms.", ECCV 2020

## Publications (Domestic)

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1. **Jaesung Rim**, Sunghyun Cho, "Per-pixel Blur Kernel Estimation Network", IPIU 2021
2. Kiyeon Kim, **Jaesung Rim**, Sunghyun Cho, "Light-weight Image Restoration Network for Image Recognition", KCC 2021

## Patents

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1. Jae Seok Choi, Manjin Kim, Sanghyun Kim, Woohyeok KIM, Insoo Kim, Min Jung Lee, **Jaesung Rim**, Deun Sol JUNG, Minsu Cho, Sunghyun Cho, " Method and apparatus with image enhancement using base image ", 12,394,084 (US, issued)
2. Sohyun Lee, **Jaesung Rim**, Boseung Jeong, Geonu Kim, ByungJu Woo, Haechan Lee, Sunghyun Cho, Suha Kwak, "Method and Apparatus for Learning Human Pose Estimation In Low-light Conditions", 10-2023-0154857 (KR, issued), 18/364,823 (US, issued)

## Professional Activities

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### Invited talk

- Realistic Blur Synthesis for Learning Image Deblurring, KCCV 2023, Poster
- Programming Practice of the State of the Arts, KCCV 2023, Tutorial
- Real-World Blur Dataset for Learning and Benchmarking Deblurring Algorithms, KCGS 2020, Poster

### Reviewer

- IEEE Computer Vision and Pattern Recognition (CVPR)
- IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)
- The Association for the Advancement of Artificial Intelligence (AAAI)
- The Visual Computer (TVC)

## Projects

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- Development of Neural Camera ISPs for Multi-Degradation Restoration and Perceptual Image Enhancement, MSIT, Korea (Mar. 2023 ~ Feb. 2026)
- eXVision: Visual Recognition in Extreme Conditions, Samsung Research Funding & Incubation Center for Future Technology, Korea (May. 2018 ~ Feb. 2026)
- Extreme Exploitation of Dark Data Research Center, MSIT, Korea (Aug. 2018 ~ Feb. 2026)
- End-to-end wide-baseline enhancement via robust alignment from raw image bursts with multiple degradation, Samsung Advanced Institute of Technology, Korea (Nov. 2022 ~ Oct. 2023)
- Real and realistic-looking synthetic data generation for low-light image enhancement and recognition, MSIT, Korea (Mar. 2020 ~ Feb. 2023)

## Teaching Experiences

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- Teaching assistance for Objective-Oriented Programming, Mar. 2023 ~ Jun. 2023
- Teaching assistance for Computer Vision (Hyundai AI Academy), Dec. 2020 ~ Dec. 2020
- Teaching assistance for Programming Practice, Mar. 2020 ~ Jun. 2020

## References

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- **Prof. Sunghyun Cho**, *Professor @ POSTECH*  
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- **Prof. Seungyong Lee**, *Professor @ POSTECH*  
Relationship: Collaborator / Mentor during the doctoral study  
Email: leesy[at]postech.ac.kr
- **Dr. Junyong Lee**, *Research Scientist @ Samsung AI Center - Toronto*  
Relationship: Collaborator / Lab colleague during the doctoral study  
Email: j.lee8[at]samsung.com